SportSathi: An AI-Based Fitness Screening and Sports Talent Guidance Platform for Schools

Project Proposal for UCS503P
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1 Higher-order goal

. . . secondary framing

This project aims to help schools move away from ad-hoc, paper-based fitness records and toward evidence-informed sports guidance, so that students with genuine athletic potential are less likely to be overlooked and students who are interested in sport receive individualised, data-backed direction rather than generic advice. The underlying goal of producing more capable, confident student-athletes is broader than any single school or sport; the immediate engineering objective is narrower: build a measurable, deployable fitness-screening and sport-suitability guidance platform that keeps a coach in control of every recommendation.

2 Time-to-value

. . . primary heuristic

- Meaningful increments will be delivered quickly by prototyping the core assessment-to-prediction workflow first – a coach records a student's general fitness scores and receives a machine-learning-generated fitness-level prediction – before layering on sport-specific tracking and progression forecasting.
- Fast feedback is prioritised over perfection: the classification model will be validated in isolation on an existing student fitness dataset before integration, so that data-quality problems surface within days rather than at the end of the semester.
- Success in the first iteration is defined by what can be shown and measured: a coach can record an assessment and receive an explainable, confidence-scored fitness prediction, gated behind mandatory coach approval before the student or parent ever sees it. Sport-specific recommendation logic and long-term progression tracking are planned for later iterations.

3 Problem Statement

Schools conduct sports activities regularly, but student fitness and performance data is rarely recorded in a structured, comparable way. Students with genuine athletic potential can go unnoticed simply because no one is systematically comparing fitness data across the school.

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Students who are interested in sport rarely receive individualised guidance on which sport suits their physical profile or what to improve, and without repeated, comparable assessments, neither the student nor the coach can reliably tell whether training is working. Coaches lack a structured way to compare students across age groups, fitness levels, or improvement needs.

This matters now because a standard, low-cost fitness-test battery (in the spirit of the Khelo India protocol) already gives schools a consistent way to measure students; the missing piece is a system that turns repeated test results into structured, coach-reviewed guidance rather than a one-time paper record.

4 Proposed Solution *. . . Software Proposition*

4.1 Overview

SportSathi will be a web-based fitness-screening and sports-guidance platform built around the following components:

- **Fitness assessment module:** records a coach-entered general fitness assessment for a student, covering strength, endurance, flexibility, speed-agility, BMI, attendance, and motivation.
- **ML-assisted fitness classification:** predicts a student's fitness category and a confidence score from the recorded assessment, using an explainable classification model.
- **Sport suitability recommendation:** recommends 2–3 suitable sports for the student's fitness profile, with a stated, plain-language reason.
- **Coach approval workflow:** holds every prediction back from students and parents until a coach explicitly approves or overrides it.
- **Student and parent guidance views:** display approved results, together with historical assessments and progress charts.
- **PDF report generation:** produces a downloadable summary report per student.

4.2 Core Workflow

A coach will record a student's general fitness assessment through the web frontend. The backend validates and stores the assessment, then invokes the machine learning service, which returns a predicted fitness level, a confidence score, and 2–3 recommended sports with stated reasons. The coach reviews the prediction on the coach dashboard and marks it approved or overridden. Only approved results become visible in the student and parent views, alongside historical assessments and a progress comparison against the student's own prior results.

4.3 Deliberate Scope Narrowing: Two-Tier Design

Collecting every conceivable metric for every possible sport, for every student, from day one would make the system too complex for a coach to realistically operate. The data model is therefore split into two tiers: **Tier 1 – General Athleticism** is a single universal fitness battery applied to every student and used to recommend suitable sports; this is the complete scope of the current semester. **Tier 2 – Sport-Specific KPIs** is a small, five-to-seven-indicator set applied only to students funnelled into a specific sport squad, together with a progression model comparing them to age-group benchmarks; this tier is explicitly treated as future scope.

4.4 Operational Constraints for an Educational, School-Access Setting

Keep the assessment-entry and dashboard interfaces usable on low-to-mid range devices via responsive web design. Avoid dependence on advanced or research-grade modelling; favour explainable, inspectable models (Decision Tree, Random Forest) over black-box architectures, since a coach must be able to trust and explain a recommendation to a parent. Design for deployability using common web hosting and a managed relational database.

5 Solution Approach *... Engineering focus*

This is an engineering course project, so the focus is on correct data modelling, honest model evaluation, and workflow automation rather than novel machine learning research.

5.1 Fitness Classification *... non-research*

Fitness-level prediction will use standard, explainable models rather than research-grade architectures: a Decision Tree and a Random Forest classifier, evaluated against each other, with class-imbalance handling (class weighting and SMOTE oversampling) investigated given the expected rarity of a top-tier fitness category in real school data. Before any accuracy figure is trusted, features that could leak the target label will be identified and excluded from the training input. The model will be surfaced to the user only as decision support: a coach must approve or override every prediction before it is acted on.

5.2 Sport Recommendation *... non-research*

Sport suitability will be a rule-based mapping from a student's fitness profile to a small set of candidate sports, rather than a learned recommendation model, keeping the logic auditable and explainable to a coach or parent.

5.3 Web Architecture *... web-based solution*

A typical three-tier web architecture will be used:

- **Frontend:** React/Vite interface for Student, Coach, Admin, and Parent roles.
- **Backend API:** Node.js/Express, handling authentication, role checks, validation, and orchestration of the ML service.
- **Machine learning service:** a Python/scikit-learn service invoked per assessment.
- **Database:** PostgreSQL, storing student profiles, assessments, coaches, recommendations, and administrative accounts in a normalised relational schema.

5.4 CI/CD and Fast Delivery

Automatically build and run backend unit tests and linting on every change; produce a repeatable deployment artefact for a staging environment; and enable safe, frequent releases of incremental features, with the Tier 1 assessment-to-prediction workflow first, and sport recommendation and reporting features after.

6 Evaluation Criteria *... measurable and attributable*

Success will be defined using measurable metrics tied to the core workflow.

6.1 Primary Evaluation Metric

Classification quality on a held-out test set: precision, recall, and macro-averaged F1-score per fitness category, with particular attention to per-class performance rather than overall accuracy alone.

Target: an explainable classifier (Decision Tree or Random Forest) that achieves at least 80% overall accuracy on a held-out split of the assessment dataset, evaluated after excluding any leaky/derived input features.

Attribution: measured directly via `scikit-learn`'s classification report and confusion matrix on the trained pipeline.

6.2 Secondary Evaluation Metrics

- **Workflow integrity:** a prediction must be verified inaccessible to the student/parent view until a coach-approval action is recorded.
- **Role isolation:** each of the four roles (Student, Coach, Admin, Parent) must access only its intended views and data.
- **Recommendation relevance:** percentage of sport recommendations a reviewing coach agrees are plausible given the student's fitness profile.
- **Coach usability:** time taken by a coach to complete a full assessment entry, kept low enough to be practical for an entire class or cohort.

6.3 Pilot Validation Plan

Seed the system with a synthetic or openly available student fitness dataset covering multiple age groups and fitness dimensions. Recruit a small number of pilot reviewers to play the Coach role, entering sample assessments and reviewing/approving generated recommendations during a short trial window. Compare the model's fitness-level predictions against the dataset's known labels, and gather simple clarity feedback on the coach dashboard from pilot reviewers.

7 Scalability *... with foresight*

Scalable in theory: the schema will be normalised around one core fact – a student is assessed repeatedly over time – so the natural unit of storage is one row per assessment, not one row per student, supporting growth in students, assessments, and concurrent usage without a redesign. The schema will be deliberately additive, so a future Tier 2 sport-specific metrics table can be introduced without modifying the Tier 1 tables.

Deployable in practice: the frontend, backend, and machine learning service will be decoupled and independently deployable, and the machine learning service will be invoked as a stateless per-request call, avoiding a persistent, hard-to-scale process.

8 Technology Stack

A mature set of tools and services supports this project:

- **Frontend framework:** React / Vite.
- **Web framework for REST APIs:** Node.js / Express.
- **Relational database:** PostgreSQL.

- **Machine learning:** Python, scikit-learn, imbalanced-learn.
- **Authentication approach:** JWT, token-based.
- **Data source:** a public/open student physical-education performance dataset, supplemented with synthetic records generated to Khelo India-style benchmark ranges.
- **Test framework:** Pytest / Jest for backend and model-evaluation testing.
- **CI runner and staging target:** GitHub Actions with a free-tier host.

These mature components let the team focus on data modelling, workflow correctness, and honest model evaluation rather than inventing new infrastructure.

9 Project Scope and Deliverables

. . . incremental delivery

9.1 Initial Deliverable (First Iteration)

- Authentication and role-based access (Student / Coach / Admin / Parent).
- Tier 1 fitness assessment entry with automatic BMI and improvement-percentage calculation.
- Trained and evaluated fitness-level classifier (leakage-checked, class-imbalance-aware).
- Coach approval/override workflow, enforced before student or parent visibility.
- CI: build, backend unit tests, and linting.
- CD to a staging environment with a single pipeline job.

9.2 Subsequent Deliverables

- Sport-suitability recommendation logic (fitness-profile-to-sport rule engine).
- Student and parent dashboards with progress charts and PDF report generation.
- Admin dashboard with school-wide distribution analytics.
- Tier 2 sport-specific KPI tracking for squad-selected students, added one sport at a time.
- A progression/forecasting model comparing a student's improvement trajectory against age-group benchmarks.

Full production-grade privacy/consent workflows, multi-school deployment, and mobile applications are acknowledged as part of the broader problem but are treated as future scope beyond this course project.

10 Risks and Mitigations

- **Data leakage / inflated accuracy:** validate that no derived/summary column is included as a model input, and report per-class metrics rather than overall accuracy alone.
- **Class imbalance:** a rare fitness category may be too small to predict reliably; mitigate with class weighting and SMOTE, and document honestly as a limitation.
- **Coach adoption:** keep assessment entry to a single form with automatic BMI calculation, and never surface an unreviewed prediction to a student or parent.

- **Data privacy / consent:** use only synthetic or openly available fitness data during development; no real student-identifying data will be collected.
- **Scope creep from sport-specific tracking:** mitigated directly by the two-tier design, which confines the current semester to a single, universal fitness battery.
- **Operational issues in the pilot:** exercise the staging environment early and keep CI/CD producing repeatable, one-command deployments.

11 Summary

This project proposes SportSathi, a deployable, coach-supervised web platform that will turn a school's fitness assessment data into structured, explainable sport-suitability guidance, without ever letting a machine learning prediction reach a student or parent unreviewed. It meets the course criteria by emphasising time-to-value through incremental delivery, planning CI/CD to support frequent and safe releases, and using measurable evaluation criteria, especially per-class classification quality and workflow integrity. The project stays focused on engineering correctness, honest evaluation, and a deliberately scoped two-tier data strategy, rather than on research-driven modelling novelty.